

ARCHITECTURE JUSTIFICATION

Peer-to-Peer Carpooling & Expense Splitter System

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Architectural Choice

We chose Layered Architecture for the Peer-to-Peer Carpooling & Expense Splitter System. The system is divided into three layers: Presentation Layer, Business Layer, and Data Layer. The Presentation Layer provides the Web/Mobile User Interface. The Business Layer contains the Order Manager, Route Compatibility, and Payment Service components. The Data Layer provides centralized storage through the Carpool Data Repository.

Reason 1 – Separation of Responsibilities

Layered Architecture separates user interaction, business logic, and data storage. This makes the system easier to understand, develop, test, and maintain. The Order Manager handles ride and request management, while the Route Compatibility Component performs route matching and validates the 1.5 km detour tolerance.

Reason 2 – Supports Carpooling Requirements

The architecture directly supports the main carpooling operations. The Order Manager manages ride creation, ride searching, ride requests, and trip confirmation. The Route Compatibility Component checks whether a driver's and passenger's routes satisfy the required detour tolerance. The Payment Service calculates and splits the fuel cost for a confirmed trip.

Security Advantage

The Data Layer provides centralized storage for users, rides, requests, trips, ratings, and verification information. The Carpool Data Repository includes encrypted storage and backups, helping protect sensitive user information.

Performance Benefit

The Route Compatibility functionality is separated into its own component, allowing route matching and detour validation to be handled independently from other ride-management operations. This supports the system's requirement for compatible carpool matches to be displayed within the specified response-time target.